

Literature Review

Many researchers have used different methods to identify product features and aspects in online customer reviews. Yang et al. [1] applied collaborative filtering and adding related search terms to find useful opinion sentences from short user queries. Although the method improved review retrieval, it was evaluated only on digital camera reviews and did not identify sentiment. Archak et al. [2] used text analysis with economic analysis to see how product features and opinions affect sales. Still, the study only focused on a few product categories and features.

Moghaddam and Ester [3] proposed a method called Opinion Digger, which is an unsupervised method that uses POS tagging, Apriori, GSP, WordNet, and KNN to find aspects and predict ratings. The method achieved good results, but it was mainly used on predefined aspects and simple filtering rules. Brody and Elhadad [4] developed an unsupervised model to find an aspect and sentiment. They used Local LDA and automatic seed generation. This model could extract aspects and sentiment without labeled data. However, it mainly used adjective-based sentiment and needed improvement for complex sentences.

Karmaker Santu et al. [5] developed GFLM to find implicit product features using language models and the EM algorithm. The method used individual words and complete sentence information. It also performed better than the baseline methods. However, this model required some explicit feature information to start the process. Zeng and Li [6] treated the task of finding implicit feature identification as a classification problem. They also used a Topic-Feature-Centroid Classifier. Their method improved feature identification by using sentence context, although it was tested only in two types of products.

Dosoula et al. [7] worked on detecting several hidden features in a single sentence with logistic regression, POS information, and feature co-occurrence. Their study showed that a single sentence can have many hidden features. However, the improvement over the baseline method was still limited. Chen et al. [8] proposed CAFE, a method that does not need labeled data. This method finds features and groups them into aspects at the same time. The method showed a better result than the baseline methods, but it may have problems with new or uncommon product words.

Bing et al. [9] introduced an HCRF-based method that did not need labeled data. It used product descriptions and customer reviews together to find popular product features. This method linked different words that referred to the same feature, but it still required some manual removal of unrelated topics. Luo et al. [10] proposed ExtRA to find important product aspects. The researchers used outside knowledge sources, word embedding methods, and clustering techniques. This method could identify important multi-word aspects, but its results depended on knowledge sources such as WordNet and Probase.

Chen et al. [11] studied hidden sentiment and found hidden aspects with BOW, Word2Vec, SVM, and CNN. The study was helpful because it focused on cases where the aspect and sentiment were not directly mentioned. But, automatically created training data labels might have errors, and some types of sentence patterns were not included. Ettaleb et al. [12] presented three weakly supervised approaches for finding aspects with CamemBERT, Guided LDA, association rules, and seed words. Their approach decreased the need for large labeled data, but it was only tested on French restaurant reviews.

Farhad et al. [13] examined different ABSA methods such as machine learning, deep learning, and transformer models. This survey showed that BERT-based methods generally give strong results, while implicit aspects, multilingual data, and low-resource languages are still challenging. Since it is a survey, however, it does not introduce a new experimental method. Gregoriades et al. [14] combined NLP, topic modeling, XGBoost, and SHAP to understand why tourists want to visit again from hotel reviews. The method provided useful explanations for hotel managers, but the study was only focused on English hotel reviews from Cyprus.

Santhiran et al. [15] introduced 41 rules based on word patterns to improve explicit feature extraction from customer feedback. Their improved rules got strong precision and recall, which showed that features are not always nouns and opinion words are not always adjectives. But, the method only focused on explicit features and did not deal with implicit or comparative features. Shen et al. [16] used sentiment networks, Graph Autoencoder, and K-means clustering to divide customers into groups based on their needs. Although the method improved customer grouping, it was only tested on one car model in China.

Mai et al. [17] worked on Vietnamese sentiment analysis with review titles, review text, PhoBERT, and emojis. The study found that titles and emojis can give useful information about feelings, but the method is designed for Vietnamese reviews and may have problems with new emojis. Habernal et al. [18] looked at implicit reasoning in arguments with crowdsourcing and neural models. The study found that AI still has problems with hidden reasoning, although this work is not directly focused on customer review feature extraction. Patil et al. [19] used ChatGPT with FastText and Logistic Regression to analyze reviews from restaurants and identify six main aspects, like food quality, service, and price. The model was applied to 4.7 million Yelp reviews, but it used only six aspects and did not perform well on some sentiment classes, mainly neutral and negative reviews.